

LAB # 7

OPEN ENDED LAB

OBJECTIVE

To apply the foundational machine learning concepts learned in Labs 1–6 to design and implement a small, custom ML project in Python.

The goal is to integrate multiple concepts learned so far into a single, working prototype that performs real-world data analysis or prediction.

PROJECT DESCRIPTION

This is an open-ended lab where you will be told individually to work on the following project ideas and build a working ML solution in Python.

Your project should demonstrate the integration of at least two machine learning models and appropriate data preprocessing, visualization, and evaluation.

PROJECT IDEAS

1. Employee Attrition Analyzer

Analyze a dataset of employee records to predict whether an employee is likely to leave the company.

Dataset: [IBM HR Analytics Employee Attrition & Performance](#)

Integration:

- Clean and preprocess categorical and numerical data.
- Train and compare **Logistic Regression**, **Decision Tree**, and **Random Forest** models.
- Evaluate results using confusion matrix, F1 score, and feature importance visualization.

2. Health Risk Classification

Design a model that predicts the likelihood of a patient having a medical condition (e.g., diabetes or heart disease) based on health indicators.

Dataset: [Pima Indians Diabetes Database](#)

•Integration:

- Perform **data preprocessing** (handle missing values, normalize numeric data).
- Apply **SVM** and **Decision Tree** for classification.
- Compare both models' results and discuss overfitting and accuracy trade-offs.

DELIVERABLES

- All source code for your project, clearly commented with Project name and roll no on top of the notebook.
- Your dataset description and preprocessing steps.
- The ML models used and how they were integrated.
- Model evaluation (metrics, confusion matrix, etc.).
- Code must be uploaded on GitHub and link paste on QOBE portal.